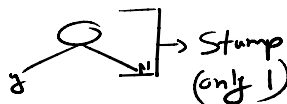


$$\text{Entropy} = -p_1 \log_2(p_1) - p_2 \log_2(p_2)$$

Small Better

Gini Impurity (Below)



Dataset

Day	Weather	Temperature	Humidity	Wind	Play Football
Day 1	Sunny	Hot	High	Weak	No
Day 2	Sunny	Hot	High	Strong	No
Day 3	Cloudy	Hot	High	Weak	Yes
Day 4	Rain	Mild	High	Weak	Yes
Day 5	Rain	Cool	Normal	Weak	Yes
Day 6	Rain	Cool	Normal	Strong	No
Day 7	Cloudy	Cool	Normal	Strong	Yes
Day 8	Sunny	Mild	High	Weak	No
Day 9	Sunny	Cool	Normal	Weak	Yes
Day 10	Rain	Mild	Normal	Weak	No
Day 11	Sunny	Mild	Normal	Strong	Yes
Day 12	Cloudy	Mild	High	Strong	Yes
Day 13	Cloudy	Hot	Normal	Weak	Yes
Day 14	Rain	Mild	High	Strong	No

1st Step

Calculate IG of Weather

Step1: Entropy of entire dataset
 $S\{+9,-5\} = -\frac{9}{14} \log_2 \frac{9}{14} - \frac{5}{14} \log_2 \frac{5}{14} = 0.94$
 Step2: Entropy of all attributes:
 • Entropy of Sunny $\{+2,-3\} = -\frac{2}{5} \log_2 \frac{2}{5} - \frac{3}{5} \log_2 \frac{3}{5} = 0.97$
 • Entropy of Cloudy $\{+4,-0\} = -\frac{4}{4} \log_2 \frac{4}{4} - \frac{0}{4} \log_2 \frac{0}{4} = 0$
 • Entropy of Rain $\{+3,-2\} = -\frac{3}{5} \log_2 \frac{3}{5} - \frac{2}{5} \log_2 \frac{2}{5} = 0.97$
 • Information Gain = Entropy(whole data) - $\frac{5}{14} \text{Ent}(S) - \frac{4}{14} \text{Ent}(C) - \frac{5}{14} \text{Ent}(R)$
 $= 0.246$

Total yes, No
 Total
 Only sunny Yes/No
 Big Better

Calculate IG of Temperature

Step1: Entropy of entire dataset
 $S\{+9,-5\} = -\frac{9}{14} \log_2 \frac{9}{14} - \frac{5}{14} \log_2 \frac{5}{14} = 0.94$
 Step2: Entropy of all attributes:
 • Entropy of Hot $\{+2,-2\} = -\frac{2}{4} \log_2 \frac{2}{4} - \frac{2}{4} \log_2 \frac{2}{4} = 1.0$
 • Entropy of Mild $\{+4,-2\} = -\frac{4}{6} \log_2 \frac{4}{6} - \frac{2}{6} \log_2 \frac{2}{6} = 0.91$
 • Entropy of Cold $\{+3,-1\} = -\frac{3}{4} \log_2 \frac{3}{4} - \frac{1}{4} \log_2 \frac{1}{4} = 0.81$
 • Information Gain = Entropy(whole data) - $\frac{4}{14} \text{Ent}(H) - \frac{6}{14} \text{Ent}(M) - \frac{4}{14} \text{Ent}(C)$
 $= 0.029$

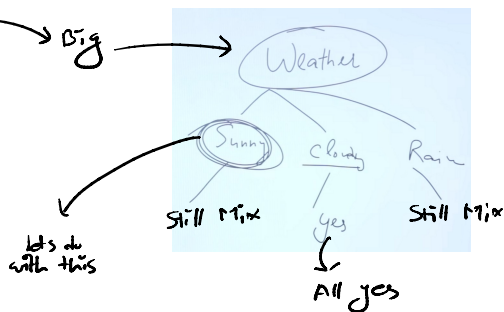
Calculate IG of Humidity

• Step1: Entropy of entire dataset
 $S\{+9,-5\} = -\frac{9}{14} \log_2 \frac{9}{14} - \frac{5}{14} \log_2 \frac{5}{14} = 0.94$
 • Step2: Entropy of all attributes:
 • Entropy of High $\{+3,-4\} = -\frac{3}{7} \log_2 \frac{3}{7} - \frac{4}{7} \log_2 \frac{4}{7} = 0.98$
 • Entropy of Normal $\{+6,-1\} = -\frac{6}{7} \log_2 \frac{6}{7} - \frac{1}{7} \log_2 \frac{1}{7} = 0.59$
 • Information Gain = Entropy(whole data) - $\frac{7}{14} \text{Ent}(H) - \frac{7}{14} \text{Ent}(N)$
 $= 0.15$

Calculate IG of Wind

• Step1: Entropy of entire dataset
 $S\{+9,-5\} = -\frac{9}{14} \log_2 \frac{9}{14} - \frac{5}{14} \log_2 \frac{5}{14} = 0.94$
 • Step2: Entropy of all attributes:
 • Entropy of Strong $\{+3,-3\} = -\frac{3}{6} \log_2 \frac{3}{6} - \frac{3}{6} \log_2 \frac{3}{6} = 1.0$
 • Entropy of Normal $\{+4,-2\} = -\frac{4}{8} \log_2 \frac{4}{8} - \frac{2}{8} \log_2 \frac{2}{8} = 0.81$
 • Information Gain = Entropy(whole data) - $\frac{6}{14} \text{Ent}(S) - \frac{8}{14} \text{Ent}(W)$
 $= 0.0478$

- Gain (S, Weather) = 0.246
- Gain (S, Temp) = 0.029
- Gain (S, Humidity) = 0.15
- Gain (S, Wind) = 0.0478



2nd Step

Only Sunny Data

Day	Weather	Temperature	Humidity	Wind	Play Football?
Day 1	Sunny	Hot	High	Weak	No
Day 2	Sunny	Hot	High	Strong	No
Day 8	Sunny	Mild	High	Weak	No
Day 9	Sunny	Cool	Normal	Weak	Yes
Day 11	Sunny	Mild	Normal	Strong	Yes

Calculate IG of Temperature

Step1: Entropy of Sunny $\{+2,-3\} = -\frac{2}{5}\log_2\frac{2}{5} - \frac{3}{5}\log_2\frac{3}{5} = 0.97$

Step2: Entropy of all attributes:

- Entropy of Hot $\{+0,-2\} = -\frac{0}{2}\log_2\frac{0}{2} - \frac{2}{2}\log_2\frac{2}{2} = 0$
- Entropy of Mild $\{+1,-1\} = -\frac{1}{2}\log_2\frac{1}{2} - \frac{1}{2}\log_2\frac{1}{2} = 1.0$
- Entropy of Cool $\{+1,-0\} = -\frac{1}{1}\log_2\frac{1}{1} - \frac{0}{1}\log_2\frac{0}{1} = 0$

Information Gain = Entropy(Sunny) - $\frac{2}{5}\text{Ent(H)} - \frac{3}{5}\text{Ent(M)} - \frac{1}{5}\text{Ent(C)}$
 $= 0.57$

Calculate IG of Humidity

Step1: Entropy of Sunny $\{+2,-3\} = -\frac{2}{5}\log_2\frac{2}{5} - \frac{3}{5}\log_2\frac{3}{5} = 0.97$

Step2: Entropy of all attributes:

- Entropy of High $\{+0,-3\} = -\frac{0}{3}\log_2\frac{0}{3} - \frac{3}{3}\log_2\frac{3}{3} = 0$
- Entropy of Normal $\{+2,-0\} = -\frac{2}{2}\log_2\frac{2}{2} - \frac{0}{2}\log_2\frac{0}{2} = 0$

Information Gain = Entropy(Sunny) - $\frac{3}{5}\text{Ent(H)} - \frac{2}{5}\text{Ent(M)}$
 $= 0.97$

Calculate IG of Wind

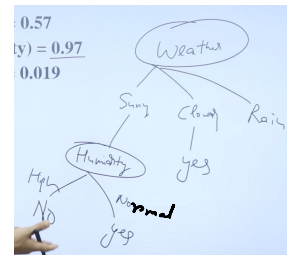
Step1: Entropy of Sunny $\{+2,-3\} = -\frac{2}{5}\log_2\frac{2}{5} - \frac{3}{5}\log_2\frac{3}{5} = 0.97$

Step2: Entropy of all attributes:

- Entropy of Strong $\{+1,-1\} = -\frac{1}{2}\log_2\frac{1}{2} - \frac{1}{2}\log_2\frac{1}{2} = 1$
- Entropy of Weak $\{+1,-2\} = -\frac{1}{3}\log_2\frac{1}{3} - \frac{2}{3}\log_2\frac{2}{3} = 0.918$

Information Gain = Entropy(Sunny) - $\frac{2}{5}\text{Ent(S)} - \frac{3}{5}\text{Ent(W)}$
 $= 0.019$

Gain (S_{sunny}, Temp) = 0.57
 Gain (S_{sunny}, Humidity) = 0.97
 Gain (S_{sunny}, Wind) = 0.019



Day	Weather	Temperature	Humidity	Wind	Play Football?
Day 4	Rain	Mild	High	Weak	Yes
Day 5	Rain	Cool	Normal	Weak	Yes
Day 6	Rain	Cool	Normal	Strong	No
	Rain	Mild	Normal	Weak	Yes
Day 14	Rain	Mild	High	Strong	No

Calculate IG of Temperature

Step1: Entropy of Rain $\{+3,-2\} = -\frac{3}{5}\log_2\frac{3}{5} - \frac{2}{5}\log_2\frac{2}{5} = 0.97$

Step2: Entropy of all attributes:

- Entropy of Hot $\{+0,-0\} = -\frac{0}{2}\log_2\frac{0}{2} - \frac{0}{2}\log_2\frac{0}{2} = 0$
- Entropy of Mild $\{+2,-1\} = -\frac{2}{3}\log_2\frac{2}{3} - \frac{1}{3}\log_2\frac{1}{3} = 0.918$
- Entropy of Cool $\{+1,-1\} = -\frac{1}{2}\log_2\frac{1}{2} - \frac{1}{2}\log_2\frac{1}{2} = 1.0$

Information Gain = Entropy(Rain) - $\frac{0}{5}\text{Ent(H)} - \frac{2}{5}\text{Ent(M)} - \frac{3}{5}\text{Ent(C)}$
 $= 0.019$

Calculate IG of Humidity

Step1: Entropy of Rain $\{+3,-2\} = -\frac{3}{5}\log_2\frac{3}{5} - \frac{2}{5}\log_2\frac{2}{5} = 0.97$

Step2: Entropy of all attributes:

- Entropy of High $\{+1,-1\} = -\frac{1}{2}\log_2\frac{1}{2} - \frac{1}{2}\log_2\frac{1}{2} = 1$
- Entropy of Normal $\{+2,-1\} = -\frac{2}{3}\log_2\frac{2}{3} - \frac{1}{3}\log_2\frac{1}{3} = 0.918$

Information Gain = Entropy(Rain) - $\frac{2}{5}\text{Ent(S)} - \frac{3}{5}\text{Ent(W)}$
 $= 0.019$

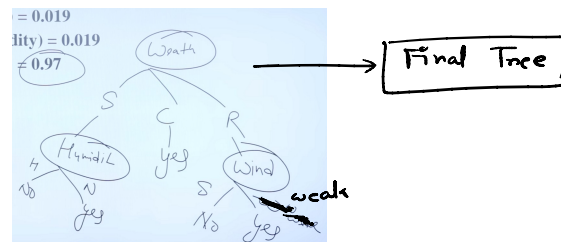
Calculate IG of Wind

Step1: Entropy of Rain $\{+3,-2\} = -\frac{3}{5}\log_2\frac{3}{5} - \frac{2}{5}\log_2\frac{2}{5} = 0.97$

Step2: Entropy of all attributes:

- Entropy of Hot $\{+0,-2\} = -\frac{0}{2}\log_2\frac{0}{2} - \frac{2}{2}\log_2\frac{2}{2} = 0$
- Entropy of Weak $\{+3,-0\} = -\frac{3}{3}\log_2\frac{3}{3} - \frac{0}{3}\log_2\frac{0}{3} = 0$

Information Gain = Entropy(Rain) - $\frac{2}{5}\text{Ent(S)} - \frac{3}{5}\text{Ent(W)}$
 $= 0.97$



11 →

Loves Popcorn	Loves Soda	Age	Loves Fruits
① Yes	Yes	7	No
② Yes	No	12	No
③ No	Yes	18	Yes
④ No	Yes	35	Yes
⑤ Yes	Yes	38	Yes
⑥ Yes	No	50	No
⑦ No	No	83	No

1st Step

Info(Before using Loves Popcorn) = $\text{Info}(3P,4N) = -(3/7)\log(3/7) - (4/7)\log(4/7) = 0.985$ → Total Dataset Entropy

Info(After using Loves Popcorn) = $(4/7)[-1/4\log(1/4) - 3/4\log(3/4)]$ → yes dataset (Fruit column)

$+(3/7)[-2/3\log(2/3) - 1/3\log(1/3)]$ → No dataset

= 0.857

Info(After using Loves Popcorn) = $(4/7) [-1/4 \log(1/4) - 3/4 \log(3/4)] \rightarrow$ yes dataset (Target column)
 $+ (3/7) [-2/3 \log(2/3) - 1/3 \log(1/3)] \rightarrow$ No dataset
 $= 0.857$

Info Gain(Loves Popcorn) = $0.985 - 0.857 = 0.128$

Info(After using Loves Soda) = $(4/7) [-3/4 \log(3/4) - 1/4 \log(1/4)]$
 $+ (3/7) [-0/3 \log(0/3) - 3/3 \log(3/3)]$
 $= 0.464$

Info Gain(Loves Soda) = $0.985 - 0.464 = 0.521$

Since Age is Numerical $\rightarrow$ Age

Info(After using Age < 36.5) = $(4/7) [-2/4 \log(2/4) - 2/4 \log(2/4)]$
 $+ (3/7) [-1/3 \log(1/3) - 2/3 \log(2/3)]$
 $= 0.965$

Info Gain(Age < 36.5) = $0.985 - 0.965 = 0.02$

We will compute information gain for other midpoints. Assume that the information gain for Age < 15 has got the highest information gain (0.291)

Age

7 12 18 35 38 50 83

9.5 15 26.5 34.5 44 46.5

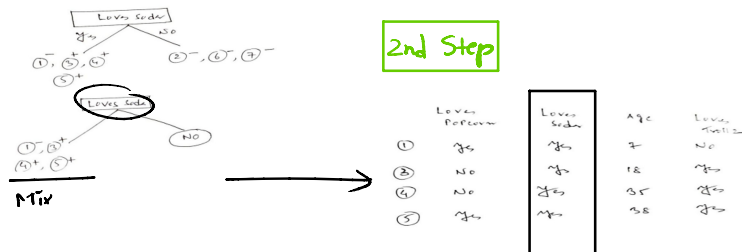
Age < 36.5

< 26.5 < 15

1, 2, 3, 4, 5, 6, 7

Finally,

Info Gain(Loves Popcorn) = 0.128
 Info Gain(Loves Soda) = 0.521
 Info Gain(Age < 15) = 0.291



Info(Before using Loves Popcorn) = Info(3P,1N) = $-(3/4) \log(3/4) - (1/4) \log(1/4) = 0.811 \rightarrow$ Total Dataset Entropy (New)

Info(After using Loves Popcorn) = $(2/4) [-1/2 \log(1/2) - 1/2 \log(1/2)]$
 $+ (2/4) [-2/2 \log(2/2) - 0/2 \log(0/2)]$
 $= 0.50$

Gain(Loves Popcorn) = $0.811 - 0.5 = 0.311$

Age

7 12 18 35 38

12.5 26.5 34.5

Info(Before using Age < 12.5) = Info(3P,1N) = $-(3/4) \log(3/4) - (1/4) \log(1/4) = 0.811$

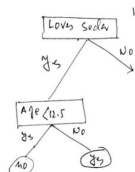
Info(After using Age < 12.5) = $(1/4) [-0/1 \log(0/1) - 1/1 \log(1/1)]$
 $+ (3/4) [-3/3 \log(3/3) - 0/3 \log(0/3)]$
 $= 0.00$

Info Gain(Age < 12.5) = $0.811 - 0.00 = 0.811$

Info gain of other mid points are less than 0.811

Entropy is the best. 1 is worst

Final Tree



Gini Impurity

$$Gini = 1 - p_{Yes}^2 - p_{No}^2$$

Weighted Gini After Split

$$p_{Yes} = \frac{\text{Number of Yes samples}}{\text{Total samples}}$$

$$p_{No} = \frac{\text{Number of No samples}}{\text{Total samples}}$$

- Gini = 0 means perfectly pure.
- Higher Gini means more mixed.

After a split, we calculate the impurity of each child node.

$$Gini_{after} = \frac{n_1}{n} Gini_1 + \frac{n_2}{n} Gini_2$$

- n = total samples
- n_1 = samples in child 1
- n_2 = samples in child 2

Rule

The best split has the lowest weighted Gini after splitting.

Entropy

Entropy

Entropy measures uncertainty.

$$Entropy = -p_{Yes} \log_2(p_{Yes}) - p_{No} \log_2(p_{No})$$

Interpretation

- Entropy = 0 means pure.
- Entropy is high when classes are mixed.

Information Gain

Information Gain tells us how much uncertainty is reduced after a split.

$$Information\ Gain = Entropy_{before} - Entropy_{after}$$

Rule

Choose the split with the highest information gain.

Higher Information Gain = Better Split

Regression Tree vs Classification Tree

Topic	Classification Tree	Regression Tree
Output	Class label	Numerical value
Example	Yes/No	Drug effectiveness
Leaf value	Majority class	Average target value
Split criterion	Gini / Entropy	SSR / MSE
Goal	Pure classes	Low prediction error

Residual and SSR

For each sample:

$$\text{Residual} = \text{Actual} - \text{Predicted}$$

Regression trees measure error using **Sum of Squared Residuals**:

$$SSR = \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

- ▶ y_i = actual value
- ▶ $\hat{y}_i$ = predicted value

Rule

Choose the split with the lowest SSR.

How Does a Regression Tree Predict?

Suppose a leaf contains drug effectiveness values:

0, 2, 3

The prediction for that leaf is the average:

$$\hat{y} = \frac{0+2+3}{3} = 1.67$$

Important

Every sample reaching this leaf will get prediction:

$$\hat{y} = 1.67$$

So

Regression tree prediction = average value in the leaf.

How Do We Choose a Split?

For classification trees, we wanted pure class groups.

For regression trees, we want groups where the numerical values are close to each other.

Good Regression Split

A good split creates child nodes where target values are similar.

Example

0, 2, 3

is a good group because values are close.

0, 100, 3

is a bad group because values are far apart.



Dosage	Gender	Drug Effectiveness
4	M	0
6	M	2
10	F	3
15	F	100
20	M	100
25	M	100
30	F	20
35	F	10
40	M	0

Testing a Categorical Split: Gender

Split the data using Gender.

Male Group

Effectiveness: 0, 2, 100, 100, 0

Female Group

Effectiveness: 3, 100, 20, 10

$$\hat{y}_M = \frac{0+2+100+100+0}{5} = 40.4 \quad \hat{y}_F = \frac{3+100+20+10}{4} = 33.25$$

$$Male = (0-40.4)^2 + (2-40.4)^2 + (100-40.4)^2 + (100-40.4)^2 + (0-40.4)^2 = 11893.2$$

$$Female = (3-33.25)^2 + (100-33.25)^2 + (20-33.25)^2 + (10-33.25)^2 = 6087$$

$$Total = 11893.2 + 6087 = 17980.2$$

male group Effectiveness: 0, 2, 100, 100, 0
 $\hat{y}_M = \frac{0+2+100+100+0}{5} = 40.4$
 female group Effectiveness: 3, 100, 20, 10
 $\hat{y}_F = \frac{3+100+20+10}{4} = 33$

$SSR(\text{Gender}) = 17929.95$

Female = $(3-33)^2 + (100-33)^2 + (20-33)^2 + (10-33)^2$
 $= 6087$
 $SSR(\text{Gender}) = 11893.2 + 6087$
 $= 17930.2$

Testing a Numerical Split: Dosage

Example Split: Dosage < 22.5

For numerical features, we test thresholds. Sorted dosages:

4, 6, 10, 15, 20, 25, 30, 35, 40

Candidate thresholds are midpoints:

5, 8, 12.5, 17.5, 22.5, 27.5, 32.5, 37.5

Example

Between dosage 20 and 25:

$\frac{20+25}{2} = 22.5$

Left Group (Dosage < 22.5)

Effectiveness: 0, 2, 3, 100, 100

$\hat{y}_L = \frac{0+2+3+100+100}{5} = 41$

Right Group (Dosage ≥ 22.5)

Effectiveness: 100, 20, 10, 0

$\hat{y}_R = \frac{100+20+10+0}{4} = 32.5$

$SSR(\text{Dosage} < 22.5) = 17883$

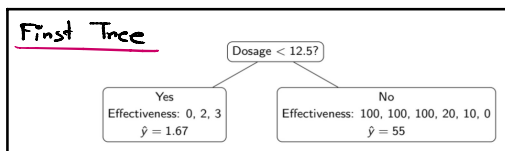
Compare All First-Level Splits

Split	SSR
Gender	17929.95
Dosage < 5	16485
Dosage < 8	14670
Dosage < 12.5	12355 → Lowest
Dosage < 17.5	17177
Dosage < 22.5	17883
Dosage < 27.5	14709
Dosage < 32.5	15374
Dosage < 37.5	16485

Best First Split
 Lowest SSR is for:

Dosage < 12.5

2-feature (12355 vs 17930.2)



$\hat{y}_{\text{left}} = \frac{0+2+3}{3} = 1.67$ $\hat{y}_{\text{right}} = \frac{100+100+100+20+10+0}{6} = 55$
 $\hat{y}_{\text{right}} = 55$

⇒ Calc Avg After Tree

Stopping Rule

We stop splitting the left branch because it has only 3 samples. In practice, regression trees use a minimum-samples rule to avoid overfitting.

The left branch has values 0, 2, 3 — these are already close. The left branch has values 0, 2, 3 and only 3 samples. Since the node is small and the values are close, we stop splitting it. The right branch has:

100, 100, 100, 20, 10, 0

This branch is still mixed, so we try to split it again.

Continue Splitting the Right Branch ⇒ (Gender SSR too high)

Candidate Splits in Right Branch

Dosage < 17.5, 22.5, 27.5, 32.5, 37.5

Choosing the Second Split

For the right branch (Dosage ≥ 12.5):

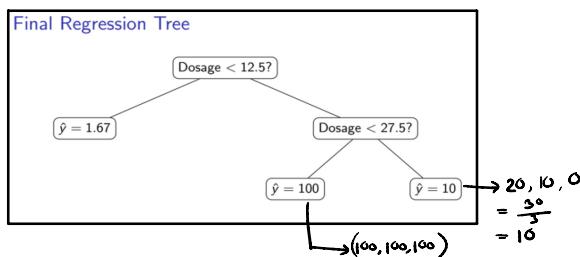
Candidate Split	SSR
Dosage < 17.5	9920
Dosage < 22.5	6275
Dosage < 27.5	200
Dosage < 32.5	7425
Dosage < 37.5	13076
Gender	10990.45

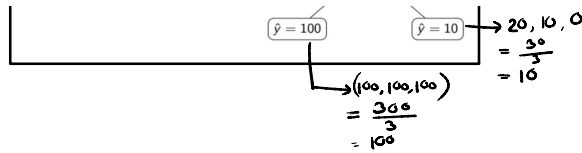
Best Second Split

Lowest SSR is:

Dosage < 27.5

Final Regression Tree





Prediction Example

Suppose a new patient receives dosage = 20.
Use the tree:

1. Is Dosage < 12.5? No.
2. Is Dosage < 27.5? Yes.

Prediction

$$\hat{y} = 100$$

The predicted drug effectiveness is **100**.

⑩ ⇒

Dose	Gender	Drug Effectiveness
5	M	10
6	M	12
7	F	20
8	F	22
20	M	80
21	M	82
22	F	95
23	F	97



Gender split
SSR is high

Candidate Split	SSR
Dose < 5.5	8805.43
Dose < 6.5	6308.00
Dose < 7.5	3822.80
Dose < 14	333.00
Dose < 20.5	3513.47
Dose < 21.5	5741.33
Dose < 22.5	8556.86
Gender	10533.00

Best Root Split

The lowest SSR is:

Dose < 14

So **Dose** becomes the root node.

Using the root split:

Dose < 14

we get two branches.

Left Branch: Dose < 14

Effectiveness values:

10, 12, 20, 22

Average prediction:

$$\hat{y} = \frac{10 + 12 + 20 + 22}{4} = 16$$

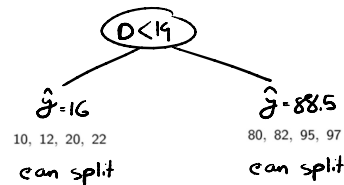
Right Branch: Dose ≥ 14

Effectiveness values:

80, 82, 95, 97

Average prediction:

$$\hat{y} = \frac{80 + 82 + 95 + 97}{4} = 88.5$$



Just checking (Not needed to show)

Step 4: Split the Left Branch

Left branch data:

Dose	Gender	Drug Effectiveness
5	M	10
6	M	12
7	F	20
8	F	22

Current prediction:

$$\hat{y} = 16$$

Current SSR:

$$(10 - 16)^2 + (12 - 16)^2 + (20 - 16)^2 + (22 - 16)^2 = 36 + 16 + 16 + 36 = 104$$

Now we test whether splitting by **Gender** improves this branch.

Step 5: Left Branch Gender Split

Split the left branch by Gender.

Male Group

Effectiveness:

10, 12

Prediction:

$$\hat{y}_M = \frac{10 + 12}{2} = 11$$

SSR:

$$(10 - 11)^2 + (12 - 11)^2 = 2$$

Female Group

Effectiveness:

20, 22

Prediction:

$$\hat{y}_F = \frac{20 + 22}{2} = 21$$

SSR:

$$(20 - 21)^2 + (22 - 21)^2 = 2$$

$$SSR(\text{Gender}) = 2 + 2 = 4$$

Decision

SSR decreases from 104 to 4, so **Gender is useful** in the left.

Just checking (Not needed to show)

Step 6: Split the Right Branch

Right branch data:

Dose	Gender	Drug Effectiveness
20	M	80
21	M	82
22	F	95
23	F	97

Current prediction:

$$\hat{y} = 88.5$$

Current SSR:

$$(80 - 88.5)^2 + (82 - 88.5)^2 + (95 - 88.5)^2 + (97 - 88.5)^2 = 72.25 + 42.25 + 42.25 + 72.25 = 229$$

Now we test whether **Gender** improves this branch.

Step 7: Right Branch Gender Split

Split the right branch by Gender.

Male Group

Effectiveness:

80, 82

Prediction:

$$\hat{y}_M = \frac{80 + 82}{2} = 81$$

SSR:

$$(80 - 81)^2 + (82 - 81)^2 = 2$$

Female Group

Effectiveness:

95, 97

Prediction:

$$\hat{y}_F = \frac{95 + 97}{2} = 96$$

SSR:

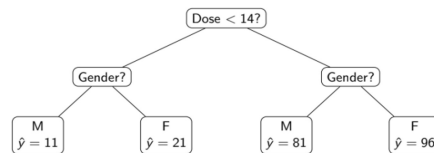
$$(95 - 96)^2 + (97 - 96)^2 = 2$$

$$SSR(\text{Gender}) = 2 + 2 = 4$$

Decision

SSR decreases from 229 to 4, so **Gender is also useful** in the right.

Final Regression Tree Using Two Features



Prediction Example

Suppose a new patient has:

Dose = 22, Gender = F

Use the tree:

1. Is Dose < 14? No.
2. Gender = F.

Prediction

$$\hat{y} = 96$$

The predicted drug effectiveness is **96**.

—

① ⇒

Day	Outlook	Temperature	Humidity	Wind	Play Tennis
Day1	Sunny	Hot	High	Weak	No
Day2	Sunny	Hot	High	Strong	No
Day3	Overcast	Hot	High	Weak	Yes
Day4	Rain	Mild	High	Weak	Yes
Day5	Rain	Cool	Normal	Weak	Yes
Day6	Rain	Cool	Normal	Strong	No
Day7	Overcast	Cool	Normal	Strong	Yes
Day8	Sunny	Mild	High	Weak	No
Day9	Sunny	Cool	Normal	Weak	Yes
Day10	Rain	Mild	Normal	Weak	Yes
Day11	Sunny	Mild	Normal	Strong	Yes
Day12	Overcast	Mild	High	Strong	Yes
Day13	Overcast	Hot	Normal	Weak	Yes
Day14	Rain	Mild	High	Strong	No

Step-1: Bootstrapping

Random

Bagging:-



Bootstrapped Dataset 1

Day	Outlook	Temperature	Humidity	Wind	Play Tennis
Day10	Rain	Mild	Normal	Weak	Yes
Day11	Sunny	Mild	Normal	Strong	Yes
Day12	Overcast	Mild	High	Strong	Yes
Day13	Overcast	Hot	Normal	Weak	Yes
Day14	Rain	Mild	High	Strong	No
Day2	Sunny	Hot	High	Strong	No

categorical
↓
Classification

DT
Gini

Random

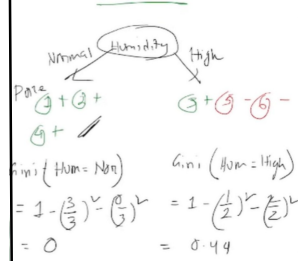
Day	Temperature	Humidity	Play Tennis
Day10	Mild	Normal	Yes
Day11	Mild	Normal	Yes
Day12	Mild	High	Yes
Day13	Hot	Normal	Yes
Day14	Mild	High	No
Day2	Hot	High	No

Temperature

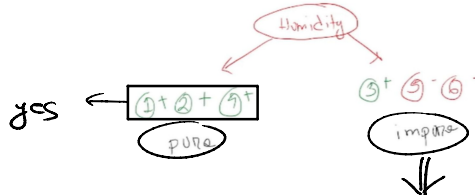


$$\begin{aligned}
 Gini(Temp = Mild) &= 1 - P(Yes)^2 - P(No)^2 \\
 &= 1 - \left(\frac{3}{4}\right)^2 - \left(\frac{1}{4}\right)^2 \\
 &= 0.375 \\
 Gini(Temp = Hot) &= 1 - P(Yes)^2 - P(No)^2 \\
 &= 1 - \left(\frac{1}{2}\right)^2 - \left(\frac{1}{2}\right)^2 \\
 &= 0.5 \\
 Gini(Temp) &= \frac{4}{6} \times 0.375 + \frac{2}{6} \times 0.5 = 0.47
 \end{aligned}$$

Humidity



$$\begin{aligned}
 Gini(Hum = Normal) &= 1 - \left(\frac{3}{5}\right)^2 - \left(\frac{2}{5}\right)^2 \\
 &= 0.44 \\
 Gini(Hum = High) &= 1 - \left(\frac{1}{2}\right)^2 - \left(\frac{1}{2}\right)^2 \\
 &= 0.5 \\
 Gini(Humidity) &= \frac{3}{6} \times 0.44 + \frac{3}{6} \times 0.5 = 0.223
 \end{aligned}$$

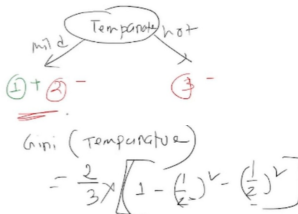
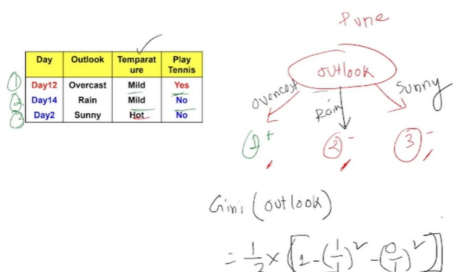


Day	Outlook	Temperature	Humidity	Wind	Play Tennis
Day10	Rain	Mild	Normal	Weak	Yes
Day11	Sunny	Mild	Normal	Strong	Yes
Day12	Overcast	Mild	High	Strong	Yes
Day13	Overcast	Hot	Normal	Weak	Yes
Day14	Rain	Mild	High	Strong	No
Day2	Sunny	Hot	High	Strong	No

Day	Outlook	Temperature	Humidity	Wind	Play Tennis
Day12	Overcast	Mild	High	Strong	Yes
Day14	Rain	Mild	High	Strong	No
Day2	Sunny	Hot	High	Strong	No

Random

Day	Outlook	Temperature	Play Tennis
Day12	Overcast	Mild	Yes
Day14	Rain	Mild	No
Day2	Sunny	Hot	No

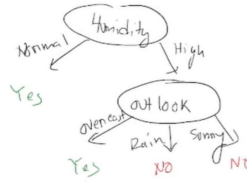


Gini (outlook)

$$= \frac{1}{3} \times \left[1 - \left(\frac{1}{3} \right)^2 - \left(\frac{2}{3} \right)^2 \right] + \frac{1}{3} \times \left[1 - \left(\frac{0}{1} \right)^2 - \left(\frac{1}{1} \right)^2 \right] + \frac{1}{3} \times \left[1 - \left(\frac{0}{1} \right)^2 - \left(\frac{1}{1} \right)^2 \right] = 0$$

Gini (temperature)

$$= \frac{2}{3} \times \left[1 - \left(\frac{1}{2} \right)^2 - \left(\frac{1}{2} \right)^2 \right] + \frac{1}{3} \times \left[1 - \left(\frac{0}{1} \right)^2 - \left(\frac{1}{1} \right)^2 \right] = 0.33$$

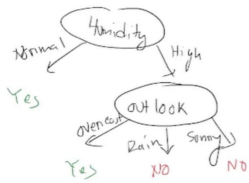


Now, more bootstrap data,

Day	Outlook	Temperature	Humidity	Wind	Play Tennis
Day10	Rain	Mild	Normal	Weak	Yes
Day11	Sunny	Mild	Normal	Strong	Yes
Day12	Overcast	Mild	High	Strong	Yes
Day13	Overcast	Hot	Normal	Weak	Yes
Day14	Rain	Mild	High	Strong	No
Day2	Sunny	Hot	High	Strong	No

Day	Outlook	Temperature	Humidity	Wind	Play Tennis
Day1	Sunny	Hot	High	Weak	No
Day2	Sunny	Hot	High	Strong	No
Day3	Overcast	Hot	High	Weak	Yes
Day4	Rain	Mild	High	Weak	Yes
Day5	Rain	Cool	Normal	Weak	Yes
Day2	Sunny	Hot	High	Strong	No

Day	Outlook	Temperature	Humidity	Wind	Play Tennis
Day6	Rain	Cool	Normal	Strong	No
Day7	Overcast	Cool	Normal	Strong	Yes
Day8	Sunny	Mild	High	Weak	No
Day9	Sunny	Cool	Normal	Weak	Yes
Day10	Rain	Mild	Normal	Weak	Yes
Day13	Overcast	Hot	Normal	Weak	Yes



Random Forest with 3 trees

Query:

Outlook	Temperature	Humidity	Wind	Play Tennis
Rain	Hot	High	Weak	No

majority voting → Yes - 1, No - 2

Finding Missing Value

⇒

Most freq. (num) → Mean/Avg (num)

Chest Pain	Good Blood Circulation	Blocked Arteries	Weight	Heart disease
No	No	No	125	No
Yes	Yes	Yes	180	Yes
Yes	Yes	No	210	No
Yes	No	??	??	No

bad Guess → Random Forest → Good guess

Chest Pain	Good Blood Circulation	Blocked Arteries	Weight	Heart disease
No	No	No	125	No
Yes	Yes	Yes	180	Yes
Yes	Yes	No	210	No
Yes	No	??	??	No

Step 1 - Initial Guess

Categorical value -

aval weight = 125 + 210 most frequent value of that

Yes	Yes	No	??	No
Yes	No	??	??	No

$$\text{Avg(weight)} = \frac{125 + 210}{2} = 167.5$$
 can do Median (But 929 will do Avg)

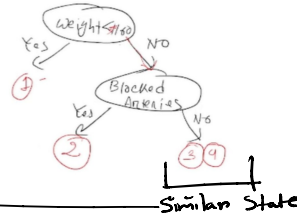
Categorical value -
 most frequent value of that
 feature corresponding to
 the target of the missing

Chest Pain	Good Blood Circulation	Blocked Arteries	Weight	Heart disease
No	No	No	125	No
Yes	Yes	Yes	180	Yes
Yes	Yes	No	210	No
Yes	No	No	167.5	No

delete 38
38% tree

Suppose 10 tree

Tree-1



Proximity Matrix - Random Forest

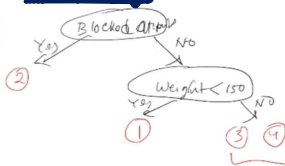
	Ex1	Ex2	Ex3	Ex4
Ex1				
Ex2				
Ex3				1
Ex4			1	

2nd Tree



	Ex1	Ex2	Ex3	Ex4
Ex1				
Ex2			1	1
Ex3		1		2
Ex4		1	2	

3rd Tree



	Ex1	Ex2	Ex3	Ex4
Ex1				
Ex2			1	1
Ex3		1		3
Ex4		1	3	

After 10 Tree

	Ex1	Ex2	Ex3	Ex4
Ex1		2	1	1
Ex2	2		1	1
Ex3	1	1		8
Ex4	1	1	8	

	Ex1	Ex2	Ex3	Ex4
Ex1		2/10	1/10	1/10
Ex2	2/10		1	1
Ex3	1/10	1/10		8/10
Ex4	1/10	1/10	8/10	

	Ex1	Ex2	Ex3	Ex4
Ex1		0.2	0.1	0.1
Ex2	0.2		0.1	0.1
Ex3	0.1	0.1		0.8
Ex4	0.1	0.1	0.8	

Updating the missing values

Chest Pain	Good Blood Circulation	Blocked Arteries	Weight	Heart disease
No	No	No	125	No
Yes	Yes	Yes	180	Yes
Yes	Yes	No	210	No
Yes	No			No

Yes No

Categorical missing value →

$$\text{Weighted Frequency of a 'Category'} = \text{Frequency of 'Category' (Data Set)} \times \text{Weight of category (Proximity Matrix)}$$

$$\Rightarrow \text{weight of 'Yes'} = \frac{\text{Proximity of 'Yes'}}{\text{All proximities}} = \frac{0.1}{0.1 + 0.1 + 0.8} = \frac{0.1}{1} = 0.1$$

$$\Rightarrow \text{weighted Frequency of 'Yes'} = \text{Frequency of 'Yes'} \times \text{weight of 'Yes'}$$

	Ex1	Ex2	Ex3	Ex4
Ex1				
Ex2				
Ex3				
Ex4				

freq = 1/3

is missing

U

All proximity

$\Rightarrow$ weighted frequency of 'Yes' = Frequency of 'Yes' $\times$ weight of 'Yes'
 $= \frac{1}{3} \times \frac{0.1}{0.1+0.1+0.8} = 0.03$

$\Rightarrow$ weight of No = $\frac{\text{Proximity of No}}{\text{All proximity}}$
 $= \frac{0.1+0.8}{0.1+0.1+0.8}$

$\Rightarrow$ weighted frequency of 'No' = Frequency of 'No' $\times$ weight of 'No'
 $= \frac{2}{3} \times \frac{0.1+0.8}{0.1+0.1+0.8} = 0.6 > 0.03$
 $\Rightarrow$ No

Yes	Yes	No	No
Yes	No		No

	Ex1	Ex2	Ex3	Ex4
Ex1		0.2	0.1	0.1
Ex2	0.2		0.1	0.1
Ex3	0.1	0.1		0.8
Ex4	0.1	0.1	0.8	

It's missing
 So, we are concerned for 4th part
 2nd is Yes

Chest Pain	Good Blood Circulation	Blocked Arteries	Weight	Heart disease
No	No	No	125	No
Yes	Yes	Yes	180	Yes
Yes	Yes	No	210	No
Yes	No	No		No

	Ex1	Ex2	Ex3	Ex4
Ex1		0.2	0.1	0.1
Ex2	0.2		0.1	0.1
Ex3	0.1	0.1		0.8
Ex4	0.1	0.1	0.8	

weighted Avg

Sample (X's) weighted average = $\frac{\text{Proximity of Sample} \times \text{Weight}}{\text{Sum of Proximity}}$

weighted Average = $125 \times \frac{0.1}{0.1+0.1+0.8} + 180 \times \frac{0.1}{0.1+0.1+0.8} + 210 \times \frac{0.8}{0.1+0.1+0.8} = 198.5$

Chest Pain	Good Blood Circulation	Blocked Arteries	Weight	Heart disease
No	No	No	125	No
Yes	Yes	Yes	180	Yes
Yes	Yes	No	210	No
Yes	No	No	198.5	No

2nd Iteration

Generate New Random trees (Maybe 10) then again update the chart

Chest Pain	Good Blood Circulation	Blocked Arteries	Weight	Heart disease
No	No	No	125	No
Yes	Yes	Yes	180	Yes
Yes	Yes	No	210	No
Yes	No	No	198.5	No

$\Rightarrow$ Again Update

$\Rightarrow$ **Stopping** when Convergence (No significant change)

Conceptual Why use proximity Matrix?

	Ex1	Ex2	Ex3	Ex4
Ex1		0.2	0.1	0.1
Ex2	0.2		0.1	0.1
Ex3	0.1	0.1		0.1
Ex4	0.1	0.1	0.8	

(3 are more connected)

Chest Pain	Good Blood Circulation	Blocked Arteries	Weight	Heart disease
No	No	No	125	No
Yes	Yes	Yes	180	Yes
Yes	Yes	No	210	No
Yes	No	??	??	No

Initial $\rightarrow$ No 167.5
1st Iteration $\rightarrow$ No 178.5 (More closer to 3rd Row)

What if we have Missing Value And No Target Value?

Testing Dataset

Chest Pain	Good Blood Circulation	Blocked Arteries	Weight	Heart disease
Yes	No	??	165	

$\Rightarrow$

IF 3/4 Values then 3/4 copies

Chest Pain	Good Blood Circulation	Blocked Arteries	Weight	Heart disease
Yes	No	??	165	

Step 1 copy of dataset based on target

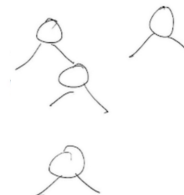
Chest Pain	Good Blood Circulation	Blocked Arteries	Weight	Heart disease
Yes	No	??	165	Yes

Chest Pain	Good Blood Circulation	Blocked Arteries	Weight	Heart disease
Yes	No	??	165	No

Step 2 Apply the process of finding missing value we learned for all copies

Step 3

Build Random Forest Classifier



After Step 2, we get the value. Then we pass data to the Tree (built by the dataset or we have this)

Chest Pain	Good Blood Circulation	Blocked Arteries	Weight	Heart disease
Yes	No	??	165	Yes

$\frac{1}{3}$ is correctly matched with tree

Chest Pain	Good Blood Circulation	Blocked Arteries	Weight	Heart disease
Yes	No	NO	165	NO

$\frac{2}{3}$ is correctly matched with tree

$\rightarrow$ We will choose this data

Regression Problem



Dataset:

Height (m)	Favourite Color	Gender	Weight (kg)
1.6	Blue	Male	88
1.6	Green	Female	76
1.5	Blue	Female	56
1.8	Red	Male	73
1.5	Green	Male	77
1.4	Blue	Female	57

Step1: Calculate (Initial) Prediction

Height (m)	Favourite Color	Gender	Weight (kg)	Initial Prediction
1.6	Blue	Male	88	71.2
1.6	Green	Female	76	71.2
1.5	Blue	Female	56	71.2
1.8	Red	Male	73	71.2
1.5	Green	Male	77	71.2
1.4	Blue	Female	57	71.2

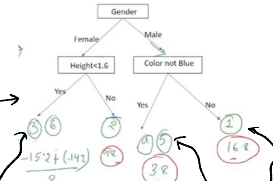
Step2: Calculate Residual

Height (m)	Favourite Color	Gender	Weight (kg)	Initial Prediction	Residual
1.6	Blue	Male	88	71.2	16.8
1.6	Green	Female	76	71.2	-4.8
1.5	Blue	Female	56	71.2	-15.2
1.8	Red	Male	73	71.2	1.8
1.5	Green	Male	77	71.2	5.8
1.4	Blue	Female	57	71.2	-14.2

Step3: Build a tree to predict the residuals

Height (m)	Favourite Color	Gender	Residuals
1.6	Blue	Male	16.8
1.6	Green	Female	-4.8
1.5	Blue	Female	-15.2
1.8	Red	Male	1.8
1.5	Green	Male	5.8
1.4	Blue	Female	-14.2

1st Tree



Tree built from current data (Most purely random). Then tree built on 1st Target (Residual Col) then Average regression/inspiration Knowledge.

Step1: Calculate (Updated) Prediction

Height (m)	Favourite Color	Gender	Weight (kg)	(Updated) Prediction
1.6	Blue	Male	88	72.9
1.6	Green	Female	76	71.6
1.5	Blue	Female	56	69.7
1.8	Red	Male	73	71.6
1.5	Green	Male	77	71.6
1.4	Blue	Female	57	69.7

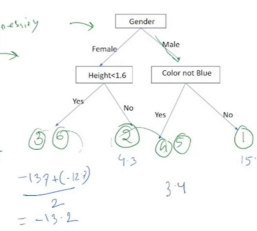
Step2: Calculate Residual

Height (m)	Favourite Color	Gender	Weight (kg)	(Updated) Prediction	Residual
1.6	Blue	Male	88	72.9	15.1
1.6	Green	Female	76	71.6	4.3
1.5	Blue	Female	56	69.7	1.9
1.8	Red	Male	73	71.6	0
1.5	Green	Male	77	71.6	0
1.4	Blue	Female	57	69.7	1

Step3: Build a tree to predict the residuals

Height (m)	Favourite Color	Gender	Residuals
1.6	Blue	Male	15.1
1.6	Green	Female	4.3
1.5	Blue	Female	1.9
1.8	Red	Male	0
1.5	Green	Male	0
1.4	Blue	Female	1

2nd Tree



$L.R = 0.1$

(after 1st iteration Updated) Prediction = Initial prediction + $L.R \times$ prediction (of Residual) from the 1st tree.

1st Ex. = $71.2 + 0.1 \times 16.8 = 72.9$
2nd Ex. = $71.2 + 0.1 \times 4.8 = 71.68$

Iteration 2

Step1: Calculate (Updated) Prediction

Height (m)	Favourite Color	Gender	Weight (kg)	(Updated) Prediction
1.6	Blue	Male	88	74.4
1.6	Green	Female	76	72.1
1.5	Blue	Female	56	69.4
1.8	Red	Male	73	71.9
1.5	Green	Male	77	71.9
1.4	Blue	Female	57	69.4

Step2: Calculate Residual

Height (m)	Favourite Color	Gender	Weight (kg)	(Updated) Prediction	Residual
1.6	Blue	Male	88	74.4	13.6
1.6	Green	Female	76	72.1	3.9
1.5	Blue	Female	56	69.4	1.5
1.8	Red	Male	73	71.9	0
1.5	Green	Male	77	71.9	0
1.4	Blue	Female	57	69.4	2.5

Step3: Build a tree to predict the residuals

Height (m)	Favourite Color	Gender	Residuals
1.6	Blue	Male	13.6
1.6	Green	Female	3.9
1.5	Blue	Female	1.5
1.8	Red	Male	0
1.5	Green	Male	0
1.4	Blue	Female	2.5

Now build a 3-regression tree to predict the latest residuals. Now prediction of an example = Initial guess + 0.1x prediction by 1-tree + 0.1x prediction by 2-tree + 0.1x prediction by 3-tree + ... until the maximum number of iteration is done or the updated residuals become insignificant.

Stopping Criteria

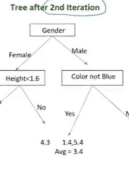
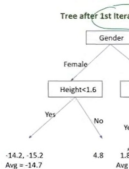
- 1 maximum Number of tree
- 2 until the Residuals become insignificant

(after 2nd iteration Updated) Prediction = Initial prediction + $L.R \times$ prediction (of Residual) from the 1st tree + $L.R \times$ prediction (of Residual) from the 2nd tree.

1st Ex. = $72.9 + 0.1 \times 15.1 = 74.4$ or $71.2 + 0.1 \times 16.8 + 0.1 \times 15.1 = 74.4$
2nd Ex. = $71.68 + 0.1 \times 4.3 = 72.1$

Prediction

Prediction from Trained model



New data

Height (m)	Favourite Color	Gender	Weight (kg)
1.7	Green	Female	?

Prediction = Initial prediction

+ $L.R \times$ prediction from 1st tree
+ $L.R \times$ " " " " " " " " " " " "
= $71.2 + 0.1 \times 4.8 + 0.1 \times 4.3$
= 72.11

Classification Problem

Classification Problem

Dataset:

Loves Popcorn	Favorite Color	Age	Loves Trail2
1.6	Blue	12	Yes
1.6	Green	87	Yes
1.5	Blue	44	No
1.8	Red	19	No
1.5	Green	32	Yes
1.4	Blue	14	Yes

Iteration 1

Step1: Calculate (Initial) Prediction

Loves Popcorn	Favorite Color	Age	Loves Trail2
1.6	Blue	12	Yes
1.6	Green	87	Yes
1.5	Blue	44	No
1.8	Red	19	No
1.5	Green	32	Yes
1.4	Blue	14	Yes

Step2: Calculate Residual

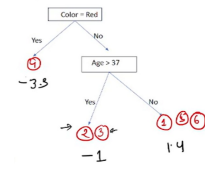
Loves Popcorn	Favorite Color	Age	Loves Trail2	(Initial) Prediction	Residual
1.6	Blue	12	Yes	0.7	0.3
1.6	Green	87	Yes	0.7	0.3
1.5	Blue	44	No	0.7	-0.3
1.8	Red	19	No	0.7	-0.3
1.5	Green	32	Yes	0.7	0.3
1.4	Blue	14	Yes	0.7	0.3

$$\text{Residual} = \text{Actual target} - \text{Prediction}$$

Step3: Build a tree to predict the residuals

Loves Popcorn	Favorite Color	Age	Loves Trail2	Residuals after initial guess
1.6	Blue	12	Yes	0.3
1.6	Green	87	Yes	0.3
1.5	Blue	44	No	-0.3
1.8	Red	19	No	-0.3
1.5	Green	32	Yes	0.3
1.4	Blue	14	Yes	0.3

Target (Residual tree)



Value for each leaf

$$= \frac{\text{Sum(Residuals)}}{\sum_{i=1}^n (\text{Previous probability}(1 - \text{Previous probability}))}$$

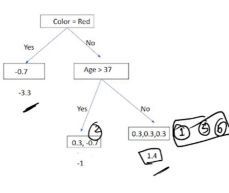
$$= \frac{0.3 + (-0.3)}{0.7(1-0.7) + 0.3(1-0.7)} = -1$$

$$= \frac{0.3 + 0.3 + 0.3}{0.7(1-0.7) + 0.3(1-0.7) + 0.3(1-0.7)} = 1.4$$

Log of odds = $\log_e \frac{p}{1-p} = \log_e \frac{P(\text{Yes}/1)}{P(\text{No}/0)} = \log_e \frac{4}{2} = \log_e 2 = \log_e \frac{\text{count(Yes)}}{\text{count(No)}} = 0.7$

Predicted Probability = $\frac{1}{1 + e^{-(\log \text{odds})}} = \frac{1}{1 + e^{-0.7}} = 0.7$

Tree after 1st Iteration



Iteration 2

Step1: Calculate (updated) Prediction

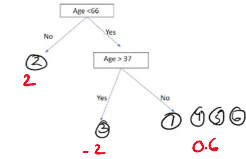
Loves Popcorn	Favorite Color	Age	Loves Trail2
1.6	Blue	12	Yes
1.6	Green	87	Yes
1.5	Blue	44	No
1.8	Red	19	No
1.5	Green	32	Yes
1.4	Blue	14	Yes

Step2: Calculate Residual

Loves Popcorn	Favorite Color	Age	Loves Trail2	(Updated) Prediction	Residual
1.6	Blue	12	Yes	0.7	0.1
1.6	Green	87	Yes	0.7	0.3
1.5	Blue	44	No	0.7	-0.3
1.8	Red	19	No	0.7	-0.3
1.5	Green	32	Yes	0.7	0.1
1.4	Blue	14	Yes	0.7	0.1

Step3: Build a tree to predict the residuals

Loves Popcorn	Favorite Color	Age	Loves Trail2	Residuals
1.6	Blue	12	Yes	0.1
1.6	Green	87	Yes	0.3
1.5	Blue	44	No	-0.3
1.8	Red	19	No	-0.3
1.5	Green	32	Yes	0.1
1.4	Blue	14	Yes	0.1



Value for each leaf

$$= \frac{\text{Sum(Residuals)}}{\sum_{i=1}^n (\text{Previous probability}(1 - \text{Previous probability}))}$$

$$= \frac{0.1 + (-0.1) + 0.1 + 0.1}{0.9(1-0.9) + 0.1(1-0.1) + 0.9(1-0.9)} = 0.6$$

Log of odds = Initial Log of odds + Learning rate X prediction (Residual) from tree-1

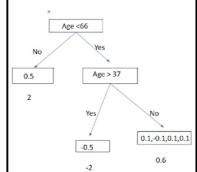
$$= 0.7 + 0.8 \times 1.4 = 1.8$$

$$= 0.7 + 0.8 \times (-1) = -0.1$$

Predicted Probability = $\frac{1}{1 + e^{-(\log \text{odds})}} = \frac{1}{1 + e^{-1.8}} = 0.9$

Predicted Probability = $\frac{1}{1 + e^{-(\log \text{odds})}} = \frac{1}{1 + e^{-0.1}} = 0.5$

Tree after 2nd Iteration



Iteration 3

Step1: Calculate (updated) Prediction

Loves Popcorn	Favorite Color	Age	Loves Trail2
1.6	Blue	12	Yes
1.6	Green	87	Yes
1.5	Blue	44	No
1.8	Red	19	No
1.5	Green	32	Yes
1.4	Blue	14	Yes

Step2: Calculate Residual

Loves Popcorn	Favorite Color	Age	Loves Trail2	Updated Prediction	Residual
1.6	Blue	12	Yes	0.9	-0.1
1.6	Green	87	Yes	0.9	-0.1
1.5	Blue	44	No	0.9	-0.1
1.8	Red	19	No	0.9	-0.1
1.5	Green	32	Yes	0.9	-0.1
1.4	Blue	14	Yes	0.9	-0.1

Step3: Build a tree to predict the residuals

Loves Popcorn	Favorite Color	Age	Loves Trail2	Residuals
1.6	Blue	12	Yes	-0.1
1.6	Green	87	Yes	-0.1
1.5	Blue	44	No	-0.1
1.8	Red	19	No	-0.1
1.5	Green	32	Yes	-0.1
1.4	Blue	14	Yes	-0.1

Can do Tree 3

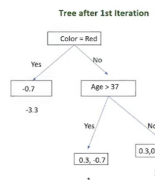
Log of odds = Initial Log of odds + L.R X Prediction (of Residual) from tree-1 + L.R X Prediction (of Residual) from Tree-2

$$= 0.7 + 0.8 \times 1.4 + 0.8 \times 0.6 = 2.3$$

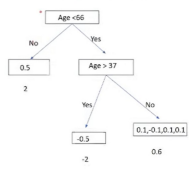
Prediction = $\frac{1}{1 + e^{-(2.3)}} = 0.91$

Prediction

Prediction from Trained model



Tree after 2nd Iteration



New data:

Loves Popcorn	Favorite Color	Age	Loves Trail2
1.2	green	45	?

→ 1.2 green 45 ? NO

Log of odds = Initial log of odd + L.R X prediction from Tree-1 + L.R X Prediction from Tree-2

$$= 0.7 + 0.8 \times (-1) + 0.8 \times (-2) = -1.7$$

Prediction = $\frac{1}{1 + e^{-(1.7)}} = 0.2 < 0.5 \rightarrow 0$